

# Yen-Hsiang (Robert) Huang

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## Educations

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**Ph.D. in Electrical and Computer Engineering | GPA: 4.0/4.0** Sept. 2022 - Present  
Georgia Tech, Atlanta, GA, USA  
**B.S. in Electrical Engineering | GPA: 4.06/4.30** Sept. 2017 - July 2022  
National Taiwan University (NTU), Taipei, Taiwan

## Research Experience

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**Graduate Research Assistant** Sept. 2022 - Present  
GTCAD Lab., Georgia Tech, Advisor: Prof. Sung Kyu Lim  
**3D IC Optimization through 3D-VIA Reduction & Inter-Die Cell Relocation** Fall 2024 - Present  
○ Timing optimization through reduced critical-path snaking and inter-die cell relocation.  
**Semiconductor Research Corporation (SRC) – JUMP 2.0 CHIMES** Fall 2023 - Spring 2024  
○ Bridged the PPA gap between legacy-node 3D ICs and advanced-node 2D ICs.  
○ Reduced WNS of legacy-node 3D ICs by 50–93% compared to SOTA, achieving 87.9–97.7% of advanced-node 2D performance.  
**Die Bonding Bumps and Pads Legalization for 3D ICs** Sept. 2022 - May 2023  
○ Developed a 3D-via legalization stage with a refinement step while minimizing runtime impact.  
○ Eliminated all 3D-via overlaps when via utilization is less than 40%.

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**Individual Study** Apr. 2020 - Sept. 2021  
EDA Lab., National Taiwan University, Advisor: Prof. Yao-Wen Chang  
**Independent Study on Physical Design**  
○ Participated in physical-design-related global contests three times across different teams and placed in the top five twice (ISPD contest 2021 and ICCAD contest 2021).  
**Individual Study** Sept. 2020 - June 2021  
ALCom Lab., National Taiwan University, Advisor: Prof. Jie-Hong Jiang  
**Independent Study on Reversible Circuit Synthesis**  
○ Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates.

## Work Experience

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**R&D Engineer** Synopsys Headquarters  
Synopsys Inc. May 2025 - Dec. 2025  
○ Investigated AI-based signal-integrity (SI) analysis for 2.5D ICs, using graph neural networks (GNNs) to predict final-stage SI performance from early-stage design data.  
○ Independently implemented two provided prototypes in C++ and merged the resulting code into the main branch.  
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**Software Engineer R&D Intern** Hsinchu, Taiwan  
Synopsys Inc. – Concurrent Clock-and-Data (CCD) Team July 2021 - Aug. 2021  
○ Enhanced the flip-flop merging algorithm in Synopsys IC Compiler II (ICC II) by considering flip-flop timing (slack).  
○ Improved timing quality in 13 real industrial benchmarks: 5.14%/3.47% in worst/total negative slack.  
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## Software Engineer R&D Intern

Cadence Inc. – Conformal Team

Hsinchu, Taiwan

June 2020 - Sept. 2020

- Provided electronic change order (ECO) test cases in a five-person team for Cadence Conformal's robustness test.
- Generated 2,000+ cases across five categories in which Conformal acted unusually and proposed three improved flows to address them.

## Publications

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- **Yen-Hsiang Huang** and Sung Kyu Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- **Yen-Hsiang Huang** and Sung Kyu Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025. [PDF]
- **Yen-Hsiang Huang**, Sai Pentapati, Anthony Agnesina, Moritz Brunion, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024.
- Sai Pentapati, Anthony Agnesina, Moritz Brunion, **Yen-Hsiang Huang**, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," ACM International Symposium on Physical Design (ISPD), 2023.

## Selected Projects

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### Mixed-Size Macro Placement and Legalization

Feb. 2021 - July 2021

- Developed a mixed-size macro placer for non-rectangular designs within a two-person team, maximizing the area for standard cells while maintaining the minimum channel needed by macros.

## Honors and Awards

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### Ministry of Science and Technology Undergraduate Research Grant

July 2021

- Topic: Wafer-Scale Physics Modeling for Finite Element Methods.

### Honorable Mention Award in the ACM ISPD contest 2021

March 2021

- 4th place worldwide in the contest.

### TSMC Scholarship for Undergraduate Students

Sept. 2020 - Jan. 2021

- Awarded to approximately 20 undergraduate students in Taiwan every semester.

### The Dean's List Award

Spring 2020

- Ranked 1st, with a 4.30/4.30 GPA in the semester.

### 4th in Data Structure and Programming Contest

March 2019

- Cadence Inc. sponsored the contest, with 125 contestants.

## Skills

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### Programming Languages

- C/C++ (expert), Python, MATLAB, LaTeX.

### Languages

- English, Mandarin (native).